

SENTINEL: A Tri-Mode Phased-Array Radar for Clutter-Resistant HAR, Vitals, and EMG-Proxy Sensing

Spatially-Enabled Nulling, Tracking, and Interleaved Networks for Edge Learning

Team Members:

- Kayoum Djedidi
- Olivier Tomé

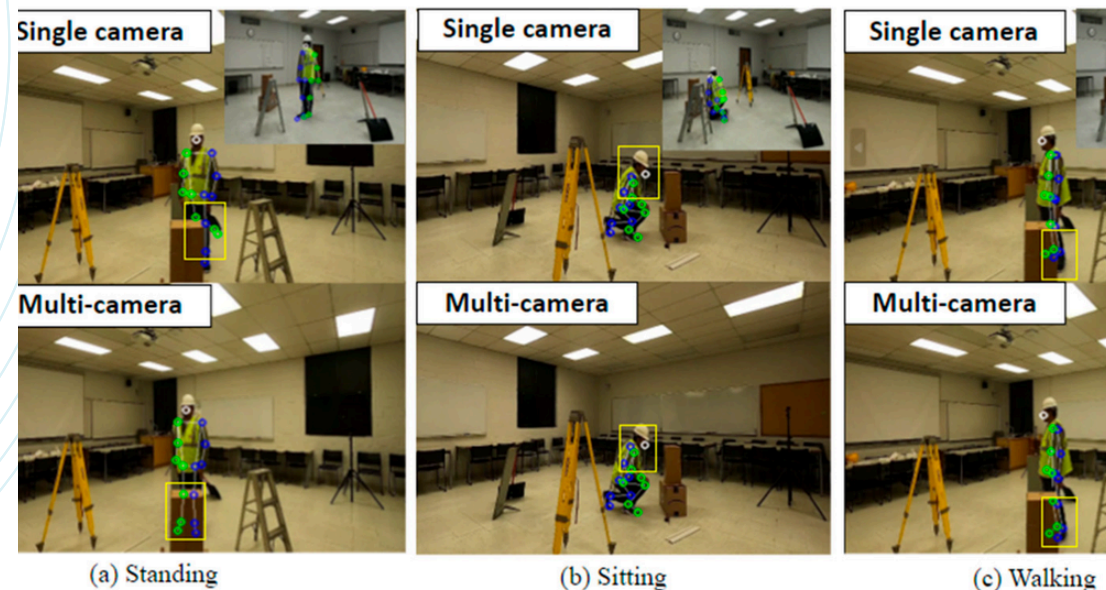
Mentor:

Prof. Tarek Djerafi

HAR = automatically recognizing what a person is doing (walk, sit, fall...) from sensor data

Camera HAR:

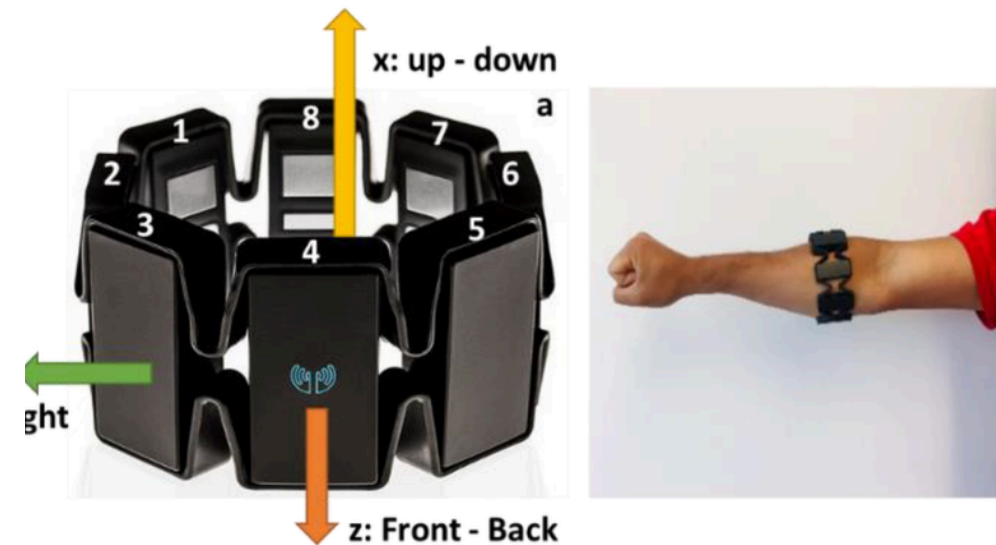
high detail but with privacy + lighting + occlusion issues



Source: [Multi-Camera-Based Human Activity Recognition for Human-Robot Collaboration in Construction](#)

Wearable HAR (IMU):

accurate but needs compliance (bulky, must wear/charge)



Source: [Data Quality and Reliability Assessment of Wearable EMG and IMU Sensor for Construction Activity](#)

Radar HAR: privacy-preserving + works in dark/through clothing, needs harder signal processing + prone to clutter



Source: [ieee-dataport: DIAT-μRadHAR: Radar micro-Doppler Signature dataset for Human Suspicious Activity Recognition](#)

This project is a tri-mode X-band phased-array radar sentinel that combines FMCW target/activity sensing, CW respiration/heartbeat extraction, CW reflectometry/myo-index sensing, and an ML classifier to move beyond single-mode HAR radar demos toward clutter-resistant, privacy-preserving health monitoring.

takeaways from SOTA & Literature review

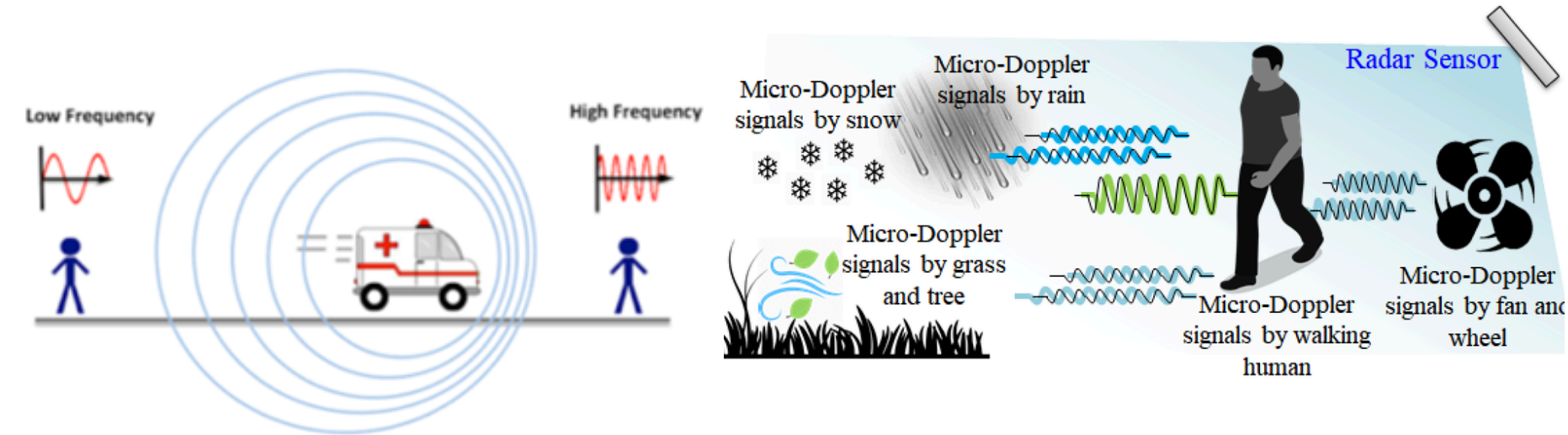
Around 14 academic sources & papers, with 10+ from 2020

- Radar enables contactless HAR + vitals using micro-Doppler/time-frequency signatures.
- **What works:**
 - Micro-Doppler spectrograms enable high HAR accuracy in controlled settings.
 - CW micro-Doppler can recover respiration/heartbeat via tiny chest motion.
 - Deep learning (CNN/RNN/Transformers) learns richer features than hand-crafted methods.
- **Gaps:**
 - Cross-environment robustness is weak (models degrade across rooms/clutter/sensor placement).
 - Most datasets are limited (often single-subject, clean lab conditions).
 - Edge constraints matter (need small models for always-on / multi-sensor deployments).
- **Key insight: efficient sequence modeling for micro-Doppler HAR**

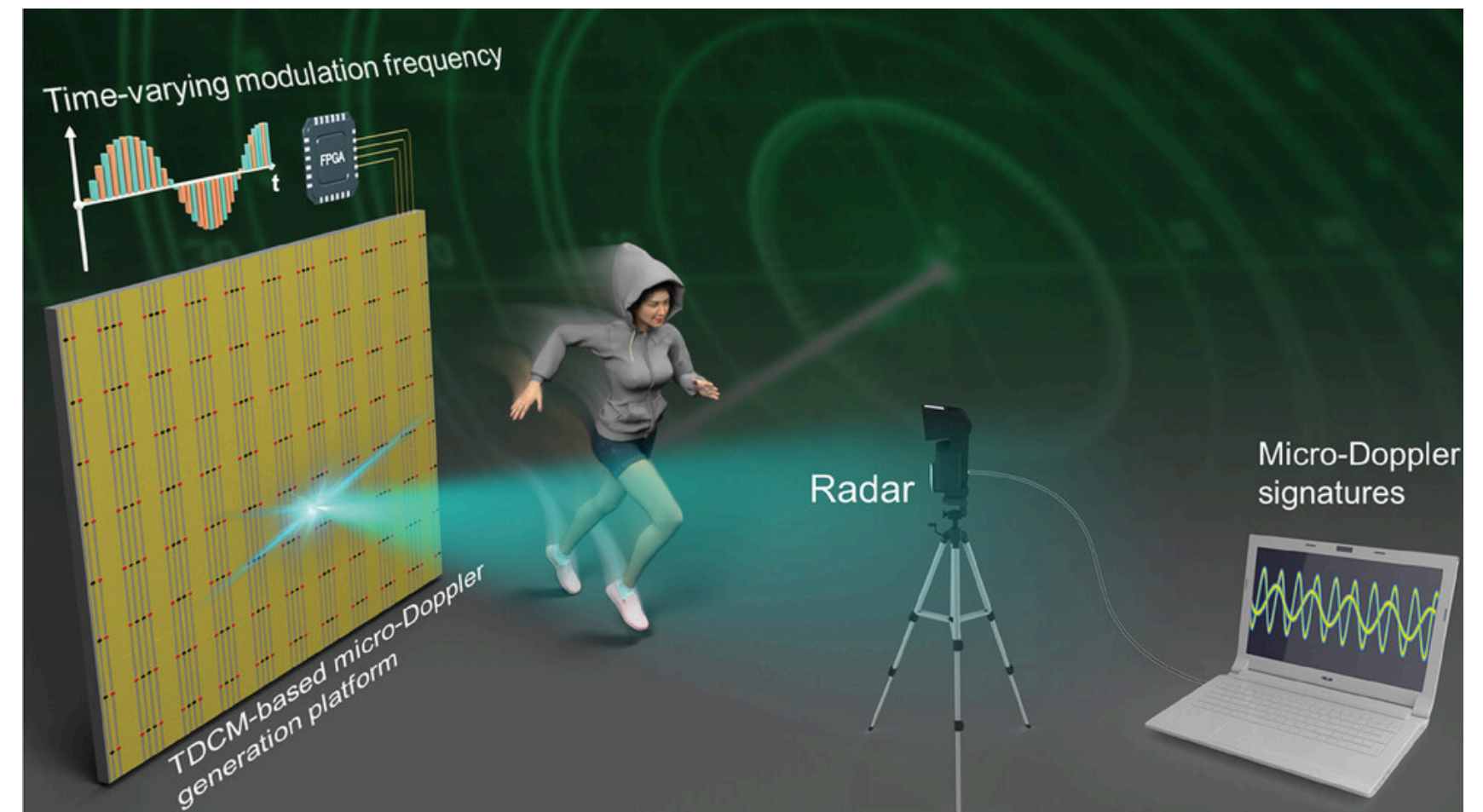
RadMamba: Efficient Human Activity Recognition through Radar-based Micro-Doppler-Oriented Mamba State-Space Model

Yizhuo Wu[✉], Student Member, IEEE, Francesco Fioranelli[✉], Senior Member, IEEE
Chang Gao[✉], Member, IEEE

- Micro-Doppler: turning motion into a 'texture' for ML



True doppler effect: Change in frequency of the sound based on approaching or withdrawal



Sources:
[Radar Micro-Doppler Signature Generation Based on Time-Domain Digital Coding Metasurface](#)
[Radar Application: Stacking Multiple Classifiers for Human Walking Detection Using Micro-Doppler](#)

MODE A (FMCW)

- detection, tracking, AoA, posture/fall cues
- Linear chirps, $B \approx 500$ MHz (0.30 m), 300 μ s ramps @ ~ 5 kHz - TDD bursts
- Tracks subjects, range-Doppler and fall cues

MODE B (CW)

- respiration + heartbeat (micro-Doppler)
- Long coherent integration
- RR: ~ 0.15 – 0.5 Hz, HR: ~ 0.8 – 2.5 Hz

MODE C (CW REFLECTOMETRY)

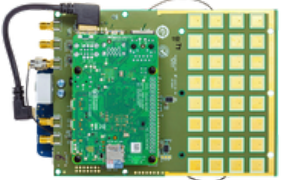
- reflectometry: muscle proxies (EMG-proxy / muscle micro-motion trends)
- Spectral energy + centroid $\rightarrow$ 'myo-index'

SCHEDULER / FUSION

- interleaves FMCW tracking and CW sensing
- locks target $\rightarrow$ runs vitals + myo window
- fuses activity, RR/HR, and myo trends

ML Model handles HAR/tracking efficiently, so signal processing budget stays available for vitals + reflectometry.

ADI CN0566
Phaser Kit



- ADAR1000 beamformer array
- ADF4159 ramp / chirp PLL
- AD9363 / PlutoSDR transceiver + Raspberry Pi 4

SMA Cable

1. Sensing Hardware

8 - 12 Ghz High
gain Vivaldi
Antenna



HB100 Frequency
reference

per-element gain /
phase calibration

Room / target
scene calibration

2. Calibration & Hardware Bring-Up Step

USB + Ethernet links

Calibrated array +
stable RF settings

Selects sensing
mode

Configures PLL /
SDR / TDD

Controls beam
steering

3. Mode Controller / Scheduler

Manual / closed-loop
target interrogation

Records IQ +
metadata

Raw IQ
recordings

Mode-tagged
metadata

Subject/demo
sessions

5. Recording & Dataset Pipeline

Future label timeline
for HAR

Screenshots +
video files

Spatial evidence: range
/ Doppler / beam angle

+

Vital evidence: RR / HR
/ phase confidence

+

Myo evidence: activation
index / centroid

6. Multi-Modal Feature Fusion

Micro-Doppler
spectrogram windows

Tiny CNN / MambaHAR model

7. ML-Ready Edge Inference

Activity / fall / risk classification

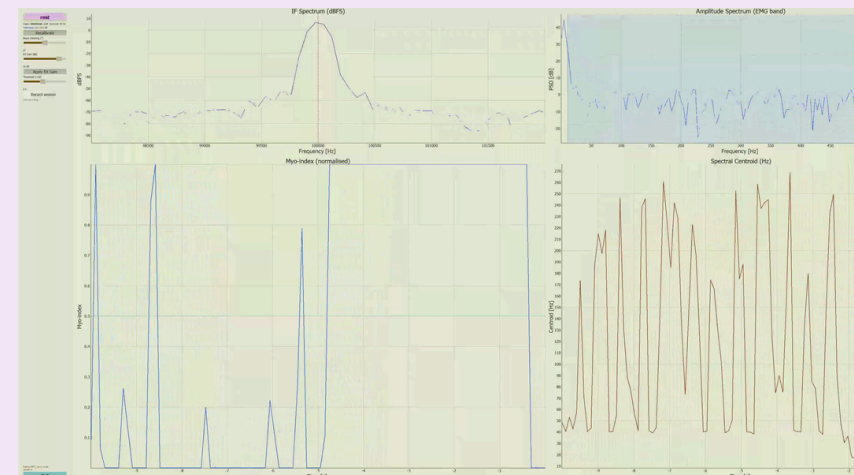
CW amplitude envelope

Spectral
energy

Activation
index

Baseline
calibration

Centroid /
band features



4C. Mode C: CW Reflectometry / Myo Proxy

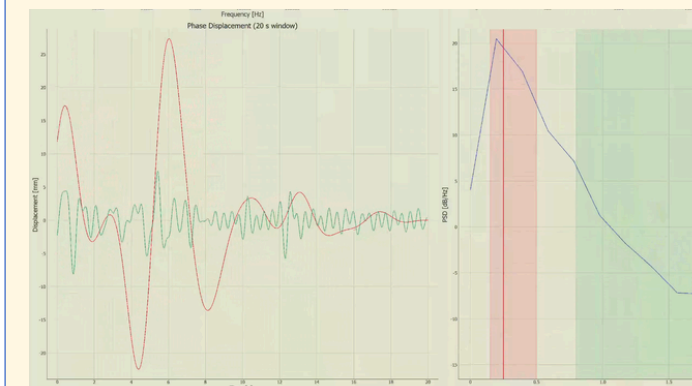
Single-tone CW IQ

Phase
unwrap

Respiration band:
 ~ 0.15 - 0.5 Hz

Chest-wall
displacement

Heartbeat band: ~ 0.8 - 2.5 Hz



4B. Mode B: CW Vitals

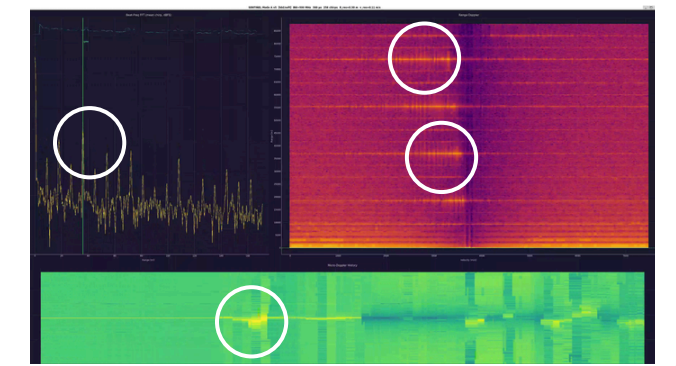
TDD soft-triggered
chirp bursts

Range FFT

MTI / clutter
suppression

Doppler FFT

Target range + motion cues

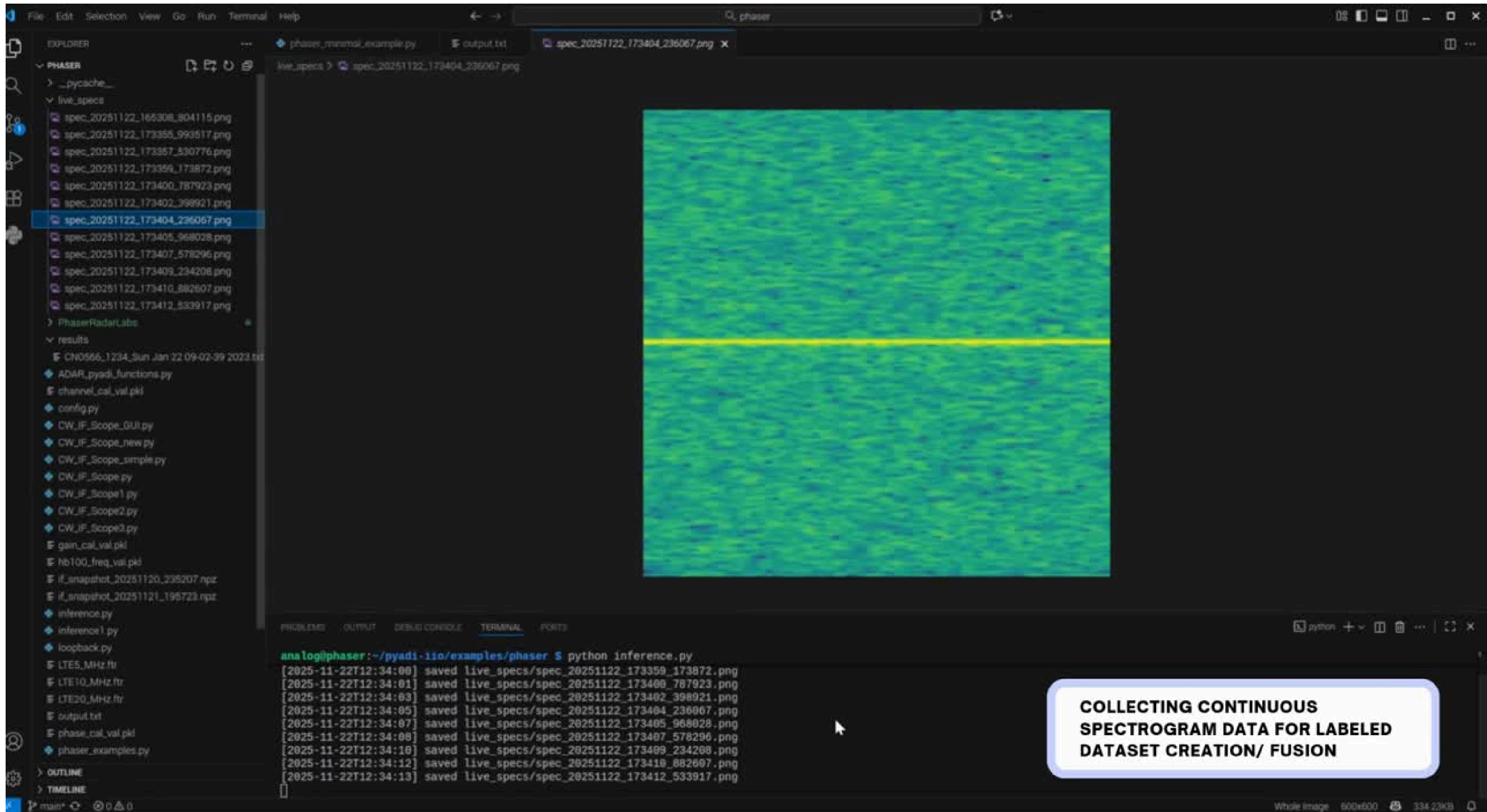


4A. Mode A: FMCW Activity Sensing

Respiration rate + heartbeat estimate

Range-Doppler map + activity / posture cues

Machine-Learning-Based Continuous Radar HAR Under Clutter



HB100 calibration cable checklist

Reference topology:

- Plug Pluto USB DATA into the Raspberry Pi attached to the Phaser.
- Unplug Pluto USB DATA from the Windows PC for this calibration.
- Keep CN0566/Phaser attached to the Raspberry Pi.
- Keep Windows PC on the same network as phaser.local.
- Expected SDR URI: ip:phaser.local:50901

HB100 source:

- Power the HB100 module from its proper supply.
- Aim it straight at the Phaser array boresight, about 0.5-2 m away.
- Keep people/moving objects away during the sweep.
- Stop all other SENTINEL/radar Python processes before running this.

Press ENTER when the cable checklist is correct and HB100 is aimed at boresight...^C

Traceback (most recent call last):
 File "/mnt/e/work/agent/workspace/SENTINEL-A-Tri-Mode-Phased-Array-Radar-for-Clutter-Resistant-HAR-Vitals-and-EMG-Proxy-Sensing/RADAR/phaser_perso/hb100_calibrate.py", line 360, in <module>
 sys.exit(main())

File "/mnt/e/work/agent/workspace/SENTINEL-A-Tri-Mode-Phased-Array-Radar-for-Clutter-Resistant-HAR-Vitals-and-EMG-Proxy-Sensing/RADAR/phaser_perso/hb100_calibrate.py", line 360, in
 input("Press ENTER when the cable checklist is correct and HB100 is aimed at boresight...")

KeyboardInterrupt

(.venv) (base) kayoum@kayoum:/mnt/e/work/agent/workspace/SENTINEL-A-Tri-Mode-Phased-Array-

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[open] reference-local: CN0566=ip:localhost Pluto=ip:phaser.local:50901

[fail] reference-rpi: Exception: No device found

[open] reference-rpi: CN0566=ip:phaser.local Pluto=ip:phaser.local:50901



**CALIBRATING PHASED ARRAY VIA
HB100, SWEEPING PEAK FREQUENCY
AND AMPLITUDE**

Conclusion: Toward a Multi-Modal Radar Health Sentinel

We demonstrated a working CN0566-based tri-mode radar stack for activity cues, respiration/heartbeat sensing, and approximate myo-index trends with a clear path toward robust ML fusion.

What We Demonstrated

- Mode A FMCW range-Doppler for target/activity cues
- Mode B CW phase sensing for respiration + heartbeat estimates
- Mode C CW reflectometry features for myo-index trends
- Calibration + recording pipeline for repeatable experiments
- Demo dashboards and subject recordings across modes

Current Limitations

- Domain shift: public datasets $\neq$ CN0566 spectrograms leakage, clutter, scaling, and hardware artifacts differ
- CI4R scale: current HDF5 split is small, train 519 / test 134 $\rightarrow$ sensitive to split and variant
- Continuous HAR not finalized, UoG20 / sequence evaluation still not fully integrated
- Metrics focus mostly on accuracy, need macro-F1, segment/event F1, and clutter robustness
- Hardware realism phaser captures show strong stationary leakage line, micro-Doppler needs better clutter removal / slow-time processing
- Physiological limits, heartbeat is chest-wall micro-motion, not ECG, and myo-index is RF proxy, not clinical EMG

Future Work

- Domain adaptation: fine-tune on 5–10 min labeled phaser data per activity
- Robustness training: Doppler shift, time warp, SNR drops, moving clutter streaks
- Continuous metrics: onset/offset-tolerant event F1 + sliding-window timeline demo
- Data alignment: standardize Fs, STFT window, normalization, and 0-Doppler notch
- Edge deployment: export TorchScript / ONNX and report Pi CPU latency + model size
- System validation: compare RR/HR/myo-index against respiration belt, ECG, and EMG